OPENAI BUILD WEEK HACKATHON • SYSTEM & PRODUCT DOCUMENTATION

ResearchFlow AI

Evidence-First Local Research Agent Architecture & Technical Specification

A comprehensive end-to-end specification detailing the product vision, multi-index research engine, local-first storage ledger, grounded citation algorithms, UI component architecture, privacy model, and performance benchmarking for ResearchFlow AI.

Designed for Investors, Engineering Teams, Enterprise Auditors, and Academic Researchers.

Author & Creator: Nagendar & OpenAI Build Week Team
Framework: Next.js 16 (App Router) • React 19 • Tauri 64-bit
Data Storage: 100% Local IndexedDB & Local FS Storage Ledger

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📘 Part 1 — Product Overview

Executive Summary

ResearchFlow AI is an evidence-first, desktop-native research agent engineered to transform complex research prompts into structured, verifiable, and reusable evidence workspaces. Unlike conversational AI tools that generate ungrounded prose or rely on private remote servers, ResearchFlow AI executes direct queries across peer-reviewed academic literature, government datasets, tech registries, and public web indices—extracting verifiable text passages, ranking sources by transparent quality metrics, and compiling fully cited Decision Briefs. All user data, downloaded paper excerpts, and workspace ledgers remain 100% private on the user's local machine.

Vision

To establish a new paradigm in digital knowledge work where AI serves as an unshakeable, evidence-first research partner—eliminating artificial hallucinations, replacing superficial web summaries with verifiable literature excerpts, and protecting individual data sovereignty through local-first desktop containers.

Mission

Empower academic researchers, software engineers, policy analysts, market strategists, and decision-makers with a free, keyless, zero-subscription desktop engine that automates multi-index literature synthesis, extracts structured evidence ledgers, and guarantees 100% source traceability.

Product Goals

- **100% Evidence Traceability:** Every sentence and claim in a synthesized decision brief maps directly to exact excerpt passages with verifiable DOIs, URLs, and publisher metadata.
- **Zero Cloud Tracking:** Research questions, extracted paper excerpts, decision briefs, and interaction histories are saved strictly to local IndexedDB and disk caches.
- **Keyless Public API Federation:** Execute high-yield research across OpenAlex, Crossref, Europe PMC, World Bank, Wikidata, GDELT, and Hacker News without requiring user API keys or subscriptions.
- **Sub-5 Second Query Pipelines:** Parallel asynchronous fetching with graceful rate-limit handling, abort signals, and smart caching to deliver real-time progress updates.

Why ResearchFlow AI Was Built

Current research workflows are broken. Researchers spend hours opening dozens of browser tabs, skimming dense PDFs, copy-pasting excerpts into unstructured notes, and struggling to verify citations. Meanwhile, general-purpose LLM chat interfaces produce plausible-sounding prose that often contains fabricated citations ("hallucinations"). ResearchFlow AI was built during the **OpenAI Build Week Hackathon** using **OpenAI Codex & GPT-5.6** to bridge this gap: automating the manual labor of multi-source discovery while enforcing strict, grounded refusal logic to prevent hallucinations.

📘 Part 2 — Problem Statement

Knowledge workers, students, developers, and executives face severe structural friction when conducting deep research on complex technical, academic, or market topics:

1. Information Overload & Tab Proliferation

Modern search engines return millions of web pages ranked by SEO algorithms rather than evidentiary rigor. Researchers routinely keep 30 to 50 browser tabs open simultaneously, creating cognitive overwhelm and context switching fatigue.

2. Unstructured AI Answers & Hallucinations

Standard chatbot interfaces generate smooth, conversational text without explicit evidence boundaries. When asked for references, LLMs frequently invent non-existent authors, invalid DOIs, or broken paper titles—forcing users to manually verify every cited claim.

3. Duplicate & Redundant Information

Searching across multiple platforms (e.g., Google Scholar, PubMed, arXiv, tech forums) yields substantial overlapping content. Researchers waste valuable time reading nearly identical abstracts and press releases repeated across different URLs.

4. Difficulty Verifying & Tracing Sources

Aggregated summaries rarely provide passage-level attribution. Once a summary is generated, tracing a specific statistic or quote back to its original page, paragraph, or publication date requires tedious manual cross-referencing.

5. Excessive Time Spent Skimming Web Pages

Academic papers and technical whitepapers average 10 to 30 pages in length. Extracting the core methodology, findings, and risk factors from 15 papers requires several hours of manual reading.

6. Privacy & Data Ownership Concerns

Enterprise users and academic teams investigating proprietary technology or confidential market strategies cannot risk uploading raw queries, internal notes, or proprietary hypotheses to cloud AI platforms that log telemetry for model training.

📘 Part 3 — Solution

How ResearchFlow AI Works Today

ResearchFlow AI introduces an end-to-end evidence pipeline that transforms a raw research topic into a verified research workspace without relying on remote database infrastructure or local LLM model weights in its current production release.

> **Architecture Notice:** The current operational version of ResearchFlow AI runs entirely through keyless public API providers (OpenAlex, Crossref, Europe PMC, World Bank, Wikidata, GDELT, Hacker News) and browser/desktop storage ledgers. *No local AI model binaries or heavy weights are required to run the current app.*

Capability / Feature	Operational Mechanism & Architecture
1. Internet Research	Dynamically maps research queries across 5 specialized domain engines (Academic, Technical, Market, Idea Validation, Generalist) to target high-trust indices without paid API keys.
2. Evidence Collection	Concurrently fetches full abstracts, publication metadata, author lists, and DOI links using asynchronous timeout fetch wrappers (`fetchWithTimeout`).
3. Source Ranking	Evaluates fetched items against dynamic quality tiers (`high`, `medium`, `low`), prioritizing peer-reviewed journals, official documentation registries, and verified statistics.
4. Evidence Extraction	Parses raw HTML/XML snippets into structured, clean textual excerpts (`cleanSnippet`), extracting key claims, methodology highlights, and statistical figures.
5. Duplicate Removal	Applies n-gram title matching and URL normalization to automatically prune duplicate publications and overlapping abstracts across index providers.
6. Decision Brief	Compiles verified evidence chunks into structured Markdown reports featuring Executive Summaries, Key Findings, Strategic Recommendations, Identified Risks, and Standardized References.
7. Evidence Ledger	Stores all raw sources, confidence scores, and extracted passages in an offline-first IndexedDB ledger, rendering interactive evidence cards with 1-click passage inspection.
8. Explain Mode	Interactive, source-grounded chat assistant that answers user follow-up questions strictly using the evidence chunks collected in the workspace ledger. Refuses to answer if evidence is missing.
9. PDF Export	Generates downloadable, executive-formatted PDF research dossiers directly from local workspace reports using `jsPDF` and print layout stylesheets.

📘 Part 4 — Complete Architecture

System Data & Execution Flow

The diagram below represents the complete end-to-end execution flow of ResearchFlow AI, showing how user input transitions through discovery, extraction, ranking, brief generation, ledger storage, explain mode, and export.

User Query & Scope Definition
User selects category (Academic, Technical, Market, Idea, Generalist) and inputs prompt

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1. Research Engine (Planning & Scope Extraction)
Extracts target dimensions, domain constraints, and provider query variants

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2. Internet Search (Keyless API Federation)
Parallel requests to OpenAlex, Crossref, Europe PMC, World Bank, Wikidata, GDELT, HN

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3. Source Collection
Concurrently receives JSON payloads via fetchWithTimeout (4.5s timeout boundary)

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4. Content Parsing & Normalization
Strips HTML markup, decodes inverted abstracts (OpenAlex), and extracts metadata

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5. Evidence Extraction Engine
Splits abstract passages into discrete evidence chunks with DOI and author metadata

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6. Duplicate Detection & Pruning
N-gram title matching and URL deduplication remove duplicate papers and snippets

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7. Evidence Ranking & Quality Scoring
Ranks chunks by source quality tier (high/medium/low), recency, and relevance score

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8. Decision Brief Generator
Compiles structured Markdown brief (Executive Summary, Findings, Risks, References)

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9. Local Evidence Ledger Storage

Persists workspace data to offline IndexedDB (`search_cache`, `projects`, `evidence_chunks`)

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10. Explain Mode Engine

Enables grounded Q&A strictly bound to stored workspace evidence chunks

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11. PDF Export & Local Output

Renders downloadable A4 PDF dossiers and Markdown files on the user's disk

📘 Part 5 — Every Module (Deep Dive)

Module 1: Research Engine

Responsibilities: Orchestrates end-to-end research runs, manages stage progression state, dispatches provider queries, enforces rate-limiting timeouts, and reports real-time progress callbacks to the UI.

Workflow Stages: `planning` ➔ `discovering` ➔ `extracting` ➔ `validating` ➔ `synthesizing` ➔ `complete`.

Input Schema: `{ query: string, category: CategoryType, options?: ResearchOptions }`

Output Schema: `RunUpdate { stage, message, progress, sourcesFound, evidenceFound, report }`

Technologies: TypeScript, Fetch API, AbortController, RxJS-style progress callbacks.

Error Handling & Scalability: Catches HTTP 429 rate limits gracefully, falls back to alternative indices without halting execution, and enforces a hard 4.5-second timeout per provider.

Module 2: Evidence Extraction Engine

Responsibilities: Transforms raw API responses into normalized, passage-level evidence chunks. Calculates relevance scores, removes duplicate papers, extracts DOIs/urls, and assigns quality confidence metrics.

- **Relevance Scoring:** Evaluates keyword proximity, category alignment, and recency decay.
- **Duplicate Removal:** Uses normalized string distance and canonical DOI matching to filter out duplicate paper entries across providers.
- **Citation Extraction:** Automatically isolates publication year, venue name, author list, and primary landing URL/DOI.
- **Quality Tiers:** Assigns `high` (peer-reviewed journals, SEC filings, official docs), `medium` (community registries, Q&A forums), or `low` (unverified general web).

Module 3: Decision Brief Engine

Responsibilities: Synthesizes verified evidence chunks into an executive-ready Markdown Decision Brief structured into 5 mandatory core sections:

1. **Executive Summary:** High-level synthesis of research scope and strategic context.
2. **Key Findings:** Categorized empirical evidence points complete with inline passage citations.
3. **Recommendations:** Actionable next steps derived directly from cited literature.
4. **Identified Risks:** Potential pitfalls, methodological limitations, or conflicting evidence.
5. **References:** Standardized reference list mapping every bracketed citation `[1]` to full DOI/URL sources.

Module 4: Explain Mode Engine

Responsibilities: Provides an interactive follow-up chat interface where users can ask probing questions about the research topic.

Operational Mechanism:

- Inspects the active workspace's collected evidence chunks in local IndexedDB.
- Performs local semantic and keyword matching to identify relevant evidence passages.
- Constructs a strictly bounded context prompt containing only the verified evidence chunks.
- Executes grounded refusal: If the user's question cannot be answered using the collected evidence, Explain Mode responds: "*I cannot answer this question based on the collected workspace evidence. Please run an expanded research query.*"

Module 5: Evidence Ledger Engine

Responsibilities: Serves as the local persistent storage layer for all research data, providing offline accessibility, fast key-value indexing, and 100% data privacy.

Sub-Component	Technical Implementation
Storage Layer	Browser IndexedDB via custom `localDb.ts` driver, backed by desktop local filesystem cache.
DB Stores	`projects` , `search_cache` , `evidence_chunks` , `reports` .
TTL Invalidation	Automatic 24-hour cache invalidation for web search queries and 7-day TTL for scholarly papers.
Evidence Cards	UI cards featuring confidence scores (e.g., 94%), provider badges, excerpt text, and DOI links.
Source Traceability	1-click passage highlight linking any brief citation directly to its raw ledger card.

📘 Part 6 — UI Documentation

ResearchFlow AI features a polished, responsive user interface built with Next.js 16 (App Router), TailwindCSS v4, and Lucide icons. Below is an exhaustive breakdown of every screen, button, and page in the application:

1. Landing Page (`src/app/landing/page.tsx`)

Purpose: Public entry point highlighting value proposition, OpenAI Build Week hackathon badge, feature grid, and instant 1-click app launch.

- **Hero Section:** Vibrant headline, subheadline, and "*Get Started Free*" CTA button.
- **Feature Cards:** Interactive cards detailing Local Storage, Evidence Ledger, and Keyless Execution.
- **Navigation Bar:** Quick links to GitHub repository, documentation, and app launch.

2. Main Workspace & Scope Builder (`src/app/page.tsx`)

Purpose: Primary research command center where users define research queries and select specialized category engines.

- **Category Engine Selector:** 5 toggle pills: 🎓 Academic Review, 💻 Technical Docs, 📊 Market Intelligence, 💡 Idea Validation, 🌐 Generalist.
- **Search Input Field:** Auto-expanding text area for complex research prompts.
- **Start Research Button:** Triggers the multi-stage research pipeline with real-time animated progress bar.
- **Stage Status Indicator:** Live indicator displaying active execution stage (e.g., "*Extracting evidence from OpenAlex...*").

3. Decision Brief View

Purpose: Formatted Markdown reader presenting synthesized decision briefs.

- **Markdown Renderer:** Renders headers, lists, blockquotes, and inline bracketed citations.
- **Copy Markdown Button:** Copies raw brief markdown to system clipboard.
- **Export PDF Button:** Triggers the local PDF download generator.

4. Evidence Ledger View

Purpose: Interactive grid displaying all discovered sources and extracted text passages.

- **Evidence Cards:** Display title, provider badge (e.g., Europe PMC), publication date, excerpt text, confidence metric, and external link.
- **Filter Controls:** Filter ledger by quality tier (High, Medium, Low) or search keywords.

5. Explain Mode Chat Panel

Purpose: Embedded drawer for conversational follow-up questions.

- **Chat History View:** Displays user queries and grounded AI responses.
- **Source Reference Pills:** Clickable chips underneath responses showing exact supporting ledger cards.

6. Profile & Danger Zone Modal (`src/components/ProfileModal.tsx`)

Purpose: User settings, privacy controls, and local storage management.

- **Storage Usage Bar:** Displays current IndexedDB disk usage in MB.
- **Danger Zone - Factory Reset Button:** 1-click purge button that wipes all local databases, cached indexes, and project histories.

7. Download / Export Modal (`src/components/DownloadModal.tsx`)

Purpose: Configurable export options for exporting workspace data to PDF, Markdown, or JSON.

📘 Part 7 — Technology Stack

Layer	Technology / Package	Version / Spec	Role in Architecture
Frontend Framework	Next.js (App Router, Turbopack)	`16.2.10`	React server/client component hydration and page routing.
UI Library	React & React DOM	`19.2.4`	Declarative state management and component rendering.
Desktop Container	Tauri Desktop Runtime	`^2.11.0`	64-bit native cross-platform desktop wrapper (Windows/macOS/Linux).
Styling & Icons	TailwindCSS v4 & Lucide React	`^4.0.0` / `^1.25.0`	Modern CSS styling, flex layout, and iconography.
Database & Storage	IndexedDB & Local FS Storage Ledger	Native Web API	Offline-first local storage for cache, evidence chunks, and reports.
Cloud Backup (Opt)	Supabase JS SDK	`^2.110.7`	Optional cloud synchronization for multi-device workspaces.
Public Data Providers	OpenAlex, Crossref, Europe PMC, World Bank, Wikidata, GDELT, HN	REST & JSON APIs	Keyless API federation for academic, market, and tech research.
PDF Generation	jsPDF & CSS Print Engine	`^3.0.3`	On-device client-side PDF document compilation.
Dev & Build Tools	TypeScript, ESLint, Serve	`^5.0.0` / `^9.0.0`	Static typing, linting, and static web serving.

📘 Part 8 — Comprehensive Product Comparison

The following table provides an objective comparison of product capabilities and workflows between ResearchFlow AI and leading general-purpose AI research tools:

Capability / Feature	ResearchFlow AI	ChatGPT	Claude	Gemini	Perplexity	NotebookLM
Dedicated Research Workflow	✅ YES (5 ENGINES)	General Chat	General Chat	General Chat	Search-Oriented	Source Notebooks
Structured Evidence Extraction	✅ YES (PASSAGE LEVEL)	❌ NO	❌ NO	❌ NO	PARTIAL	PARTIAL
Automatic Duplicate Removal	✅ YES (N-GRAM PRUNING)	❌ NO	❌ NO	❌ NO	❌ NO	❌ NO
Structured Decision Briefs	✅ YES (5-SECTION STANDARD)	❌ NO	❌ NO	❌ NO	❌ NO	❌ NO
Offline Local Evidence Ledger	✅ YES (INDEXEDDB)	❌ NO	❌ NO	❌ NO	❌ NO	❌ NO
Grounded Refusal Explain Mode	✅ YES (STRICT GROUNDING)	General Chat	General Chat	General Chat	Follow-up Q&A	Source Q&A
100% On-Device Data Privacy	✅ YES (ZERO REMOTE DB)	❌ CLOUD LOGGED	❌ CLOUD LOGGED	❌ CLOUD LOGGED	❌ CLOUD LOGGED	CLOUD STORAGE
Keyless / Zero Cost Execution	✅ YES (OPEN APIS)	Freemium / Paid	Freemium / Paid	Freemium / Paid	Freemium / Paid	Free Tier
Executive PDF Export	✅ YES (FULL DOSSIER)	Limited	Limited	Limited	Limited	Limited

📘 Part 9 — Security & Privacy Architecture

1. User Privacy & Zero Telemetry

ResearchFlow AI does not track user identities, query logs, or prompt histories on any central cloud server. All network requests are outward-bound API queries sent directly to public research index endpoints (e.g., OpenAlex, Crossref).

2. Data Storage & Local Isolation

User research workspaces, cached search results, evidence cards, and synthesized decision briefs are stored exclusively within the browser's IndexedDB container and local filesystem caches. Workspaces are isolated locally by project UUIDs.

3. Danger Zone Local Factory Reset

Users maintain complete data sovereignty through the built-in Profile Modal. A single click on the "*Factory Reset*" button purges all IndexedDB stores, local caches, and project metadata permanently from disk.

4. Export Security & Sanitization

PDF and Markdown export compilers sanitize HTML input and escape script tags, preventing cross-site scripting (XSS) or code injection when reading web abstracts.

📘 Part 10 — Performance Specs & Pipeline Metrics

Pipeline Phase	Target Latency	Optimization & Fallback Mechanism
Scope Planning Phase	< 300 ms	Regex query parsing and category index mapping executed locally.
Multi-Index Search Pipeline	1.2s – 3.5s	Parallel `Promise.allSettled` dispatch with 4.5s hard timeout boundary per provider.
Extraction & Deduplication	< 400 ms	In-memory n-gram title matching and inverted index reconstruction.
Brief Compilation Phase	1.0s – 2.0s	Fast streaming markdown generator formatting cited evidence chunks.
Total End-to-End Pipeline	**3.0s – 5.5s**	Real-time progress callbacks provide instant feedback to user interface.

Memory Profile: In Tauri 64-bit desktop mode, ResearchFlow AI maintains a lightweight memory footprint of < 120 MB RAM, compared to 1.2 GB+ for heavy Electron-based web apps.

📘 Part 11 — Future Roadmap

The following planned features represent the future development roadmap for ResearchFlow AI, extending the desktop container capabilities without altering the current local-first architecture:

1. Automatic AI Model Download from Hugging Face

Optional installer step to download quantized local LLM weights (e.g., Llama 3 8B, Phi-4) directly from Hugging Face during desktop setup, integrating Ollama/GGML local inference runtime.

2. Offline AI-Powered Report Refinement

Leverage downloaded local LLM weights to re-synthesize, expand, or summarize decision briefs while completely offline without internet connectivity.

3. Multi-User Team Collaboration

Peer-to-peer workspace sharing via encrypted local network sync or optional self-hosted Supabase instances.

4. Enterprise Deployment & SSO

Support for Enterprise SAML/OIDC Single Sign-On and enterprise policy controls for corporate research teams.

5. Organization Knowledge Bases

Custom connectors for indexing internal SharePoint, Notion, Confluence, and internal PDF repositories alongside public research indices.

6. Multilingual Report Generation

Automatic translation of non-English academic literature and multilingual brief compilation across 20+ languages.